

$$\sqrt{c^2 - b^2}$$



$$a^2 + b^2 = c^2$$

QUANTITATIVE APTITUDE

$$\int \sin x \, dx = -\cos x + c$$

$$\int \frac{1}{\sqrt{1-x^2}} \, dx = \arcsin x + c$$

$$\int \frac{1}{\sqrt{1-x^2}} \, dx = -\arccos x + c$$

$$\int x^n \, dx = \frac{x^{n+1}}{(n+1)} + c$$

$$\int \frac{1}{\cos^2 x} \, dx = \tan x + c$$

$$\int x^n \, dx = \frac{x^{n+1}}{(n+1)} + c$$

$$\int a^x \, dx = \frac{a^x}{\ln a} + c$$

Number System

- Detailed Theory Notes
- Covered Each Subtopic
- Quick Revision

HELPFUL FOR ALL COMPETITIVE EXAMS



Number System

Introduction: A number system is a mathematical notation used to represent numbers. It provides a set of rules and symbols for expressing quantities and performing arithmetic operations. Different number systems exist, each with its own base or radix, which determines the number of unique symbols used and their positional value. The most commonly used number systems are the decimal system (base 10) and binary system (base 2). Here we will study only the decimal system (base 10).

Indian and International Numeral System: In the Indian numeral system, every number is represented by using ones, thousands, lakhs and crores, while in the international numeral system, every number is represented by ones, thousands, millions, billions, trillions etc.

Indian Numeral System Table:

Periods	Places	Numerical Value	Number of Digits
Ones	Ones	1	1
	Tens	10	2
	Hundreds	100	3
Thousands	Thousand	1000	4
	Ten Thousands	10000	5
Lakhs	Lakhs	100000	6
	Ten Lakhs	1000000	7
Crores	Crores	10000000	8
	Ten Crores	100000000	9



International Numeral System Table:

Periods	Places	Numerical Value	Number of Digits
Ones	Ones	1	1
	Tens	10	2
	Hundreds	100	3
Thousands	Thousands	1000	4
	Ten Thousands	10000	5
	Hundred Thousands	100000	6
Millions	Millions	1000000	7
	Ten Millions	10000000	8
	Hundred Millions	100000000	9
Billions	Billions	1000000000	10
	Ten Billions	10000000000	11
	Hundred Billions	100000000000	12

Place Value and Face Value: The face value of a digit in a number is the value of the digit itself. The place value of a digit in a number is the product of that digit and the numerical value of the place at which the digit occurs.

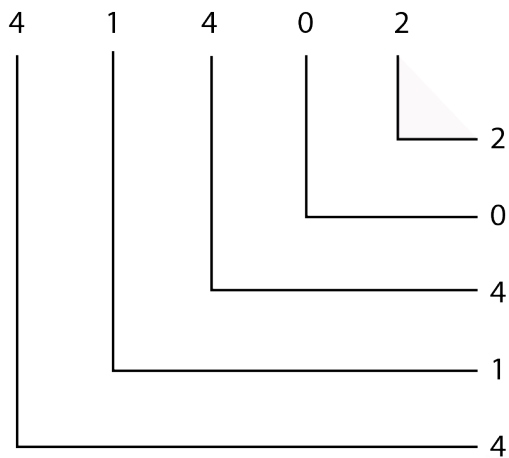
Important Points:

- The face value of a digit remains the same no matter where it occurs.
- The face value and place value of digit 0 is always 0.
- Place Value = Face value $\times$ Numerical Value of place.

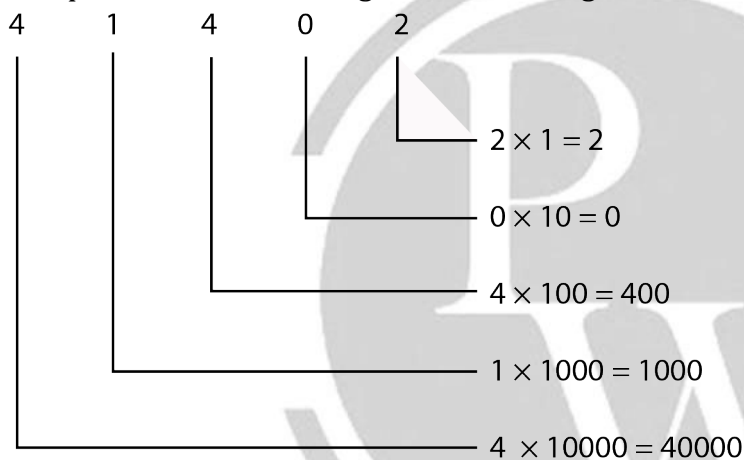


Example: The face value and place value of each digit in the number 41402 are:

Solution: The face value of each digit in 41402 are given as:



The place value of each digit in 41402 are given as:



Example: Find the difference between the place value and face value of 5 in the number 145278.

Solution: The place value of 5 in 145278 = 5000

The face value of 5 in 145278 = 5

Required difference = $5000 - 5 = 4995$

Types of Numbers: There are various types of numbers in mathematics, each with its own unique properties and characteristics. Some of the most common types of numbers include:

Natural Numbers (N): Natural numbers, also known as counting numbers, are the set of positive integers starting from 1 and continuing indefinitely. In mathematical notation, they are represented as $N = \{1, 2, 3, 4, \dots\}$.

Whole Numbers (W): Whole numbers are similar to natural numbers, but they include zero as well. In mathematical notation, they are represented as $W = \{0, 1, 2, 3, \dots\}$.



Integers (Z): Integers are a set of whole numbers and their negative counterparts, along with zero. In mathematical notation, they are represented as $Z = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$.

Rational Numbers (Q): Rational numbers are numbers that can be expressed as a fraction p/q , where p and q are integers and q is not equal to zero. The decimal expansion of a rational number is either terminating or recurring decimal.

For example, $1/2$, -3 , 0.75 , and $-0.333\dots$ are all rational numbers.

Irrational Numbers: Irrational numbers are those numbers which cannot be expressed as a fraction p/q , where p and q are integers and $q \neq 0$. Irrational numbers have non-repeating, non-terminating decimal expansions.

For example, $\sqrt{2}$, $\sqrt{5}$, π , e , $5.020020002\dots$ are all irrational numbers.

Real Numbers (R): Real numbers encompass all rational and irrational numbers. They include all possible points on the number line, both positive and negative, integers, fractions, and the infinite variety of decimals.

To summarize:

Natural Numbers (N): 1, 2, 3, ...

Whole Numbers (W): 0, 1, 2, ...

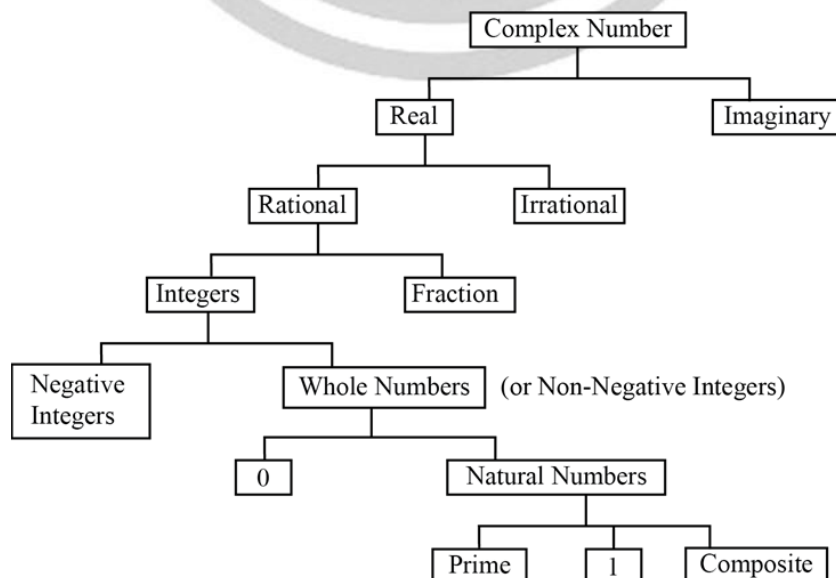
Integers (Z): $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

Rational Numbers (Q): Fractions and integers.

Irrational Numbers: Non-repeating, non-terminating decimals.

Real Numbers (R): All possible points on the number line.

Number Tree: The classification of numbers can be understood by the following diagram known as number tree.





Prime Numbers: Those numbers who have exactly two factors are called prime numbers.

Prime Numbers upto 100: {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97}

Method to Find Whether a Number N is Prime or Not:

Step 1: Find the square root of N. Suppose it is m.

Step 2: List down all the prime numbers less than equal to m.

Step 3: If N is not divisible by any of these numbers then N will be the prime number.

Example: Check Whether the following numbers are prime or not?

(i) 299 (ii) 307

Solution:

(i) $\sqrt{299} = 17.3$

The prime numbers less than 17.3 are: 2, 3, 5, 7, 11, 13, 17.

Now, out of these numbers 299 is divisible by 13.

Hence, 299 is not a prime number.

(ii) $\sqrt{307} = 17.5$

The prime numbers less than 17.5 are: 2, 3, 5, 7, 11, 13, 17.

Now, 307 is not divisible by any of the above numbers.

Hence, 307 is a prime number.

Composite numbers: Those numbers who have more than two factors are called composite numbers.

Even Numbers: Those numbers which are integer multiple of 2 are called even numbers.

OR

Those numbers whose unit digit is any one of 0, 2, 4, 6, 8 are called even numbers.

{..., -6, -4, -2, 0, 2, 4, 6, ...} is the set of even numbers.

Odd Numbers: Those numbers which are not integer multiple of 2 are called odd numbers.

OR

Those numbers whose unit digit is any one of 1, 3, 5, 7, 9 are called even numbers.

{..., -5, -3, -1, 1, 3, 5, ...} is the set of odd numbers.



Divisibility Rule:

Divisibility Rule for 2: A number is divisible by 2 if its last digit is even, i.e., it ends in 0, 2, 4, 6, or 8.

Divisibility Rule for 3: A number is divisible by 3 if the sum of its digits is divisible by 3.

Divisibility Rule for 4: A number is divisible by 4 if the number formed by its last two digits is divisible by 4.

Divisibility Rule for 5: A number is divisible by 5 if its last digit is either 0 or 5.

Divisibility Rule for 6: A number is divisible by 6 if it is divisible by both 2 and 3 (it follows from the rules for divisibility by 2 and 3).

Divisibility Rule for 7: This one is a bit trickier. Take the last digit of the number, double it, and subtract the result from the remaining part of the number. If the result is divisible by 7 (including 0), then the original number is also divisible by 7.

Divisibility Rule for 8: A number is divisible by 8 if the number formed by its last three digits is divisible by 8.

Divisibility Rule for 9: A number is divisible by 9 if the sum of its digits is divisible by 9.

Divisibility Rule for 10: A number is divisible by 10 if it ends in 0.

Divisibility Rule for 11: Take the alternating sum of the digits (subtract the first digit from the second, then add the third digit, subtract the fourth digit, and so on). If the result is divisible by 11 (including 0), then the original number is also divisible by 11.

Example: If the number 45217a43 is divisible by 11, then find the value of a.

Solution: We know that a number is divisible by 11 if the difference between the sum of digits at odd and even places is either 0 or a multiple of 11.

Sum of digits at odd places = $4 + 2 + 7 + 4 = 17$

Sum of digits at even places = $5 + 1 + a + 3 = 9 + a$

Difference = $(9 + a) - 17 = a - 8$

$\Rightarrow a - 8 = 0$

$\Rightarrow a = 8.$

Example: If the number 341x2851y is divisible by 72, then find the value of x + y.

Solution: A number is divisible by 72 if it is divisible by 8 and 9 both.

341x2851y is divisible by 8 if its last three digits are divisible by 8.

Now, 51y is divisible by 8 if $y = 2$



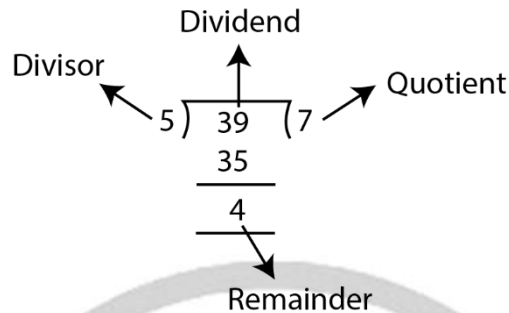
Now, $341x28512$ is divisible by 9 if the sum of its digits is divisible by 9.

The sum of digits of $341x28512 = 3 + 4 + 1 + x + 2 + 8 + 5 + 1 + 2 = 26 + x$

Now, $26 + x$ is divisible by 9 if $x = 1$.

Therefore $x + y = 1 + 2 = 3$

Division Lemma: Suppose we have to divide 39 by 5.



$$\text{Dividend} = \text{Divisor} \times \text{Quotient} + \text{Remainder}$$

Example: In a division sum, the divisor is 7 times the quotient and 3 times the remainder. If the remainder is 56, what is the dividend?

Solution: Remainder = 56

$$\text{Divisor} = 56 \times 3 = 168$$

$$\text{Quotient} = \frac{168}{7} = 24$$

$$\begin{aligned}
 \text{Dividend} &= \text{Divisor} \times \text{Quotient} + \text{Remainder} \\
 &= 168 \times 24 + 56 = 4032 + 56 = 4088
 \end{aligned}$$

Example: If a number is divided by 306, the remainder is 78. If the same number is divided by 18, then What is the remainder?

Solution: Let the number be N.

If a number is divided by 306, the remainder is 78.

In this case, let the quotient be m.

$$\text{Then, } N = 306m + 78 \quad [\text{Dividend} = \text{Divisor} \times \text{Quotient} + \text{Remainder}]$$

$$N = 306m + 72 + 6$$

$$N = 18 \times 17m + 18 \times 4 + 6$$

$$N = 18(17m + 4) + 6$$

Clearly, when N is divisible by 18, the remainder is 6.

Short Trick:

Since 18 divides 306 completely. So divide 78 directly by 18 and in this case the remainder is 6.
Hence, the required remainder is 6.



Unit digit:

I. Suppose we have to find the unit digit of the expression $(634 \times 441 \times 276 \times 188 \times 193)$.

In this case, we multiply only the unit digits of the given numbers,

$$\text{Now, } 4 \times 1 \times 6 \times 8 \times 3 = 576$$

So, the unit digit of the given expression is 6.

II. If we have to find the unit digit of 3^{83} .

In this case note down the unit digits of 3^n for $n = 1, 2, 3, \dots$

$$3^1 = 3$$

$$3^2 = 9$$

$$3^3 = 7$$

$$3^4 = 1$$

$$3^5 = 3$$

....

Here, the unit of 3^n repeats after every 4th power.

So, divide 83 by 4 and we get the remainder 3.

Hence, unit digit of $3^{83} = \text{unit digit of } 3^3 = \text{unit digit of } 27 = 7$.

Formulae for the unit digit of a^n where a is any digit and n is a natural number:

The unit digit of $1^n = 1$ for every n

$$\text{The unit digit of } 2^n = \begin{cases} 2, & \text{if } n = 4k + 1 \\ 4, & \text{if } n = 4k + 2 \\ 8, & \text{if } n = 4k + 3 \\ 6, & \text{if } n = 4k \end{cases}$$

$$\text{The unit digit of } 3^n = \begin{cases} 3, & \text{if } n = 4k + 1 \\ 9, & \text{if } n = 4k + 2 \\ 7, & \text{if } n = 4k + 3 \\ 1, & \text{if } n = 4k \end{cases}$$

$$\text{The unit digit of } 4^n = \begin{cases} 4, & \text{if } n = 2k + 1 \\ 6, & \text{if } n = 2k \end{cases}$$

The unit digit of $5^n = 5$ for every n

The unit digit of $6^n = 6$ for every n

$$\text{The unit digit of } 7^n = \begin{cases} 7, & \text{if } n = 4k + 1 \\ 9, & \text{if } n = 4k + 2 \\ 3, & \text{if } n = 4k + 3 \\ 1, & \text{if } n = 4k \end{cases}$$



$$\text{The unit digit of } 8^n = \begin{cases} 8, & \text{if } n = 4k + 1 \\ 4, & \text{if } n = 4k + 2 \\ 2, & \text{if } n = 4k + 3 \\ 6, & \text{if } n = 4k \end{cases}$$

$$\text{The unit digit of } 9^n = \begin{cases} 9, & \text{if } n = 2k + 1 \\ 1, & \text{if } n = 2k \end{cases}$$

The unit digit of $0^n = 0$ for every n

Example: Find the unit digit of $(37)^{27} + (54)^{36} + (32)^{56}$

Solution: $27 = 4 \times 6 + 3 = 4k + 3$

The unit digit of $(37)^{27} =$ the unit digit of $(7)^{27} = 3$

36 is an even number

The unit digit of $(54)^{36} =$ the unit digit of $(4)^{36} = 6$

$56 = 4 \times 14 = 4k$

The unit digit of $(32)^{56} =$ the unit digit of $(2)^{56} = 6$

The unit digit of $(37)^{27} + (54)^{36} + (32)^{56} =$ the unit digit of $(3+6+6=15) = 5$

Factorial: The factorial of a natural number is equal to the product of the first n natural numbers and it is denoted by $N!$.

$$N! = 1 \times 2 \times 3 \times \dots \times N$$

Example: $5! = 1 \times 2 \times 3 \times 4 \times 5 = 120$

Maximum Value of m for Which p^m Divides $N!$: Let p be a prime number. Then, the maximum value of m for which p^m divides $N!$ completely $= \left[\frac{N}{p} \right] + \left[\frac{N}{p^2} \right] + \left[\frac{N}{p^3} \right] + \dots$

where $[x]$ denotes the greatest integer less than equal to x .

Example: Find the maximum value of n for which 7^n divides $400!$

Solution: The maximum value of n for which 7^n divides $400!$

$$\begin{aligned} &= \left[\frac{400}{7} \right] + \left[\frac{400}{7^2} \right] + \left[\frac{400}{7^3} \right] + \dots \\ &= \left[\frac{400}{7} \right] + \left[\frac{400}{49} \right] + \left[\frac{400}{343} \right] + \dots \\ &= [57.14] + [8.16] + [1.16] + \dots \\ &= 57 + 8 + 1 = 66 \end{aligned}$$



Short Tricks:

We successively divide 400 by 7 until we get the quotient zero.

Divisor	Dividend	Quotient
7	400	57
7	57	8
7	8	1
7	1	0

Maximum value of n for which p^m divides $N!$ = $57 + 8 + 1 = 66$

Number of Zeros in $N!$: The multiplication of 2 and 5 yields a zero. So, to find the number of zeros in $N!$, we find the number of 2 and 5 in the prime factorisation of $N!$. Since the number of 5 is always smaller than the number of 2 in the prime factorisation of $N!$ so, the number of zeros at the end of $N!$ is equal to the number of 5 in the prime factorisation of $N!$

Example: Find the number of zeros at the end of $300!$

Solution:

Divisor	Dividend	Quotient
5	300	60
5	60	12
5	12	2
5	2	0

Required number of zeros = $60 + 12 + 2 = 74$

Number of Factors: If the prime factorization of a natural number N is

$$N = p_1^{q_1} p_2^{q_2} p_3^{q_3} \dots$$

- I. The number of factors of $N = (q_1 + 1)(q_2 + 1)(q_3 + 1) \dots$
- II. The number of factors of N which are multiples of $p_1 = (q_1)(q_2 + 1)(q_3 + 1) \dots$

Example: Find the number of factors of 1800.



Solution: $1800 = 2^3 \times 3^2 \times 5^2$

Number of factors of 1800 = $(3 + 1)(2 + 1)(2 + 1) = 4 \times 3 \times 3 = 36$

Example: Find the number of even factors of 1800.

Solution: $1800 = 2^3 \times 3^2 \times 5^2$

Number of even factors of 1800 = $(3)(2 + 1)(2 + 1) = 3 \times 3 \times 3 = 27$

Example: Find the number of factors of 1800 which are multiples of 5..

Solution: $1800 = 2^3 \times 3^2 \times 5^2$

Number of factors of 1800 which are multiples of 5 = $(3 + 1)(2 + 1)(2) = 4 \times 3 \times 2 = 24$

Sum of Factors: If the prime factorization of a natural number N is given by

$$N = p_1^{q_1} p_2^{q_2} p_3^{q_3} \dots$$

Then, the sum of all the factors of N

$$= (1 + p_1^1 + p_1^2 + \dots + p_1^{q_1})(1 + p_2^1 + p_2^2 + \dots + p_2^{q_2})(1 + p_3^1 + p_3^2 + \dots + p_3^{q_3}) \dots$$

Example: Find the sum of all the factors of 720.

Solution: $720 = 2^4 \times 3^2 \times 5^1$

$$\begin{aligned} \text{Sum of all the factors of 720} &= (1 + 2^1 + 2^2 + 2^3 + 2^4)(1 + 3^1 + 3^2)(1 + 5^1) \\ &= (1 + 2 + 4 + 8 + 16)(1 + 3 + 9)(1 + 5) \\ &= 31 \times 13 \times 6 = 2418 \end{aligned}$$